PROJECTION FORMULA FOR LOWER SHRIEK ALONG LOCALLY QUASI-FINITE MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: the checks are asserted as immediate rather than spelled out, but each sits on fully displayed data. (1) Compatibility of the pure tensor rule with the colimit transition maps and with restriction mappings (proof lines 2080-2082) is manifest from the displayed rule, which keeps the same support Z and applies the linear pullback to s. (2) Compatibility with the stalk identifications of lemma-lqf-f-shriek-stalk (proof lines 2082-2084) is exhibited by the displayed stalk diagram, whose lower horizontal arrow is 'the obvious isomorphism' (proof line 2101) between the fibre direct sums. No verbatim 'we omit' occurs in the proof.

Allowed standard prerequisites (editorial).

- Definition of lower shriek for locally quasi-finite morphisms (Definition definition-f-shriek-lqf, canonical tag 0F6R). cited at proof lines 2041-2044. f_!F is the sheafification of the presheaf f_{p!}F, which is what the construction works on.
- Presheaf support colimit description (Section section-finite-support, canonical tag 0F6E). cited at proof lines 2054-2061. Gives f_{p!}F(V) = colimit over Z of H_Z(F), the description on which the pure tensor rule is defined.
- Stalk description of locally quasi-finite lower shriek (Lemma lemma-lqf-f-shriek-stalk, canonical tag 0F5F). cited at proof lines 2082-2084 and 2098-2100. Identifies the stalks of both sides with fibre direct sums; the stalk of can becomes the obvious isomorphism.

Source reference. Original reference 0F6R: sources/stacks-chow/tags.txt, source lines 18981–18981. Cited at proof lines 2041-2044: f_!F is the sheafification of the presheaf f_{p!}F, reducing the construction to the presheaf level.

Source reference. Original lemma 0F5F: sources/stacks-chow/tags.txt, source lines 18935–18935. Cited at proof lines 2082-2084 and 2098-2100: the vertical arrows of the stalk diagram, identifying both sides with the fibre direct sums.

Source reference. Original reference 0F6E: sources/stacks-chow/tags.txt, source lines 18969–18969. Cited at proof lines 2054-2061: the description f_{p!}F(V) = colim_Z H_Z(F) on which the pure tensor rule is defined.

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Source reference. Original reference 03EK: sources/stacks-chow/tags.txt, source lines 4459–4459. Cited at proof lines 2051-2053: the presheaf tensor product used in the intermediate map.

Lemma 0.2. *Let $f : X \to Y$ be a locally quasi-finite morphism of schemes. Let Λ be a ring. Let $\mathcal{F}$ be a sheaf of Λ-modules on $X_{étale}$ and let $\mathcal{G}$ be a sheaf of Λ-modues on $Y_{étale}$. There is a canonical isomorphism*

$$can : f_!\mathcal{F} \otimes_\Lambda \mathcal{G} \longrightarrow f_!(\mathcal{F} \otimes_\Lambda f^{-1}\mathcal{G})$$

of sheaves of Λ-modules on $Y_{étale}$.

Proof. Recall that $f_!\mathcal{F} = (f_{p!}\mathcal{F})^\#$ by Definition 0F6R where $f_{p!}\mathcal{F}$ is the presheaf constructed in Section 0F6E. Thus in order to construct the arrow it suffices to construct a map

$$f_{p!}\mathcal{F} \otimes_{p,\Lambda} \mathcal{G} \longrightarrow f_{p!}(\mathcal{F} \otimes_\Lambda f^{-1}\mathcal{G})$$

of presheaves on $Y_{étale}$. Here the symbol $\otimes_{p,\Lambda}$ denotes the presheaf tensor product, see Modules on Sites, Section 03EK. Let V be an object of $Y_{étale}$. Recall that

$$f_{p!}\mathcal{F}(V) = \operatorname{colim}_Z H_Z(\mathcal{F}) \quad \text{and} \quad f_{p!}(\mathcal{F} \otimes_\Lambda f^{-1}\mathcal{G})(V) = \operatorname{colim}_Z H_Z(\mathcal{F} \otimes_\Lambda f^{-1}\mathcal{G})$$

See Section 0F6E. Our map will be defined on pure tensors by the rule

$$(Z, s) \otimes t \longmapsto (Z, s \otimes f^{-1}t)$$

(for notation see below) and extended by linearity to all of $(f_{p!}\mathcal{F} \otimes_{p,\Lambda} \mathcal{G})(V) = f_{p!}\mathcal{F}(V) \otimes_\Lambda \mathcal{G}(V)$. Here the notation used is as follows

(1) $Z \subset X_V$ is a locally closed subscheme finite over V,
(2) $s \in H_Z(\mathcal{F})$ which means that $s \in \mathcal{F}(U)$ with $\operatorname{Supp}(s) \subset Z$ for some $U \subset X_V$ open such that $Z \subset U$ is closed, and
(3) $t \in \mathcal{G}(V)$ with image $f^{-1}t \in f^{-1}\mathcal{G}(U)$.

Since the support of $s \in \mathcal{F}(U)$ is contained in Z it is clear that the support of $s \otimes f^{-1}t$ is contained in Z as well. Thus considering the pair $(Z, s \otimes f^{-1}t)$ makes sense. It is immediate that the construction commutes with the transition maps in the colimit $\operatorname{colim}_Z H_Z(\mathcal{F})$ and that it is compatible with restriction mappings. Finally, it is equally clear that the construction is compatible with the identifications of stalks of $f_!$ in Lemma 0F5F. In other words, the map *can* we've produced on stalks at a geometric point $\overline{y}$ fits into a commutative diagram

$$\begin{array}{ccc} (f_!\mathcal{F} \otimes_\Lambda \mathcal{G})_{\overline{y}} & \xrightarrow[can_{\overline{y}}]{} & f_!(\mathcal{F} \otimes_\Lambda f^{-1}\mathcal{G})_{\overline{y}} \\ \downarrow & & \downarrow \\ (\bigoplus \mathcal{F}_{\overline{x}}) \otimes_\Lambda \mathcal{G}_{\overline{y}} & \longrightarrow & \bigoplus (\mathcal{F}_{\overline{x}} \otimes_\Lambda \mathcal{G}_{\overline{y}}) \end{array}$$

where the direct sums are over the geometric points $\overline{x}$ lying over $\overline{y}$, where the vertical arrows are the identifications of Lemma 0F5F, and where the lower horizontal arrow is the obvious isomorphism. We conclude that *can* is an isomorphism as desired. □